Memorization and Generalization under Training-Label Randomization on Tabular Data

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***Abstract*—Inspired by Zhang et al.'s random-label experiments, we study memorization and generalization in multiclass tabular classification. Using the UCI Dry Bean dataset, we compare logistic regression (LR), random forest (RF), and a multilayer perceptron (MLP) under five training-label resampling probabilities and three random seeds. Our notebook has completed data cleaning, exploratory data analysis (EDA), feature-selection comparisons, 18 clean-label tuning trials, and 45 main experimental fits. Clean-test macro-F1 generally decreases with label randomization. At full resampling, RF still fits 84.61% of noisy training labels on average, but its clean-test macro-F1 is only 0.1222. We summarize our preliminary findings, limitations, team contributions, and remaining project work.**

I. Assigned Paper and Project Objectives

Our assigned paper is *Understanding Deep Learning Requires Rethinking Generalization* by Zhang et al. (ICLR 2017) [1]. It shows that large neural networks can fit randomly assigned training labels, even when these labels contain no useful relationship to input images. This raises a basic question: does successful training indicate useful learning or merely fitting the observed labels?

Rather than reproducing the image experiments, we investigate numerical features in a seven-class bean dataset and compare three model families.

We ask two main research questions:

RQ1: How does training-label randomization affect clean-test macro-F1 across LR, RF, and MLP?

RQ2: Is a model's ability to fit randomized training labels associated with its performance on untouched test labels?

Our goal is to compare training fit with generalization, rather than to rank machine learning algorithms universally.

II. Dataset, EDA, and Preprocessing

A. Data Audit and Class Imbalance

We downloaded the Dry Bean dataset from the UCI Machine Learning Repository [2], based on measurements described by Koklu and Ozkan [3].

The original dataset contains 13,611 observations, 16 numerical image-derived size and shape predictors, and seven target classes. Examples of the features include Area, Perimeter, Compactness, and ShapeFactor2.

Our data audit found no missing values and 68 exact duplicate rows. We removed duplicates before splitting the dataset to reduce the risk of identical observations appearing in both training and test sets.

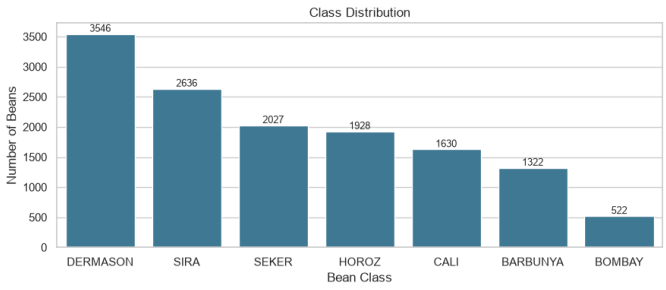


Fig. 1. Original Dry Bean class distribution.

After cleaning, 13,543 distinct observations remained.

The original class distribution is imbalanced (Fig. 1). DERMASON contains 3,546 observations, while BOMBAY contains only 522.

Because overall accuracy may hide weak performance on smaller classes, we use macro-F1 as our primary evaluation metric. Accuracy and balanced accuracy are also recorded.

B. Initial Feature Findings

Scatter plots of Area versus Perimeter and Compactness versus ShapeFactor2 show both class structure and overlap.

Area and Perimeter are strongly correlated, with Pearson correlation $r \approx 0.967$.

This suggests that some geometric predictors may contain similar information. However, correlation alone does not justify removing a feature.

We therefore compare selecting the 10 highest-ranked features using ANOVA `SelectKBest` with retaining all 16 features.

In our current tuning results, all three selected configurations retain all 16 features.

After deduplication, two stratified splits (random state 17) produced 9,480 training, 2,031 validation, and 2,032 test observations, approximately a 70/15/15 split.

Standardization for LR and MLP is performed inside pipelines fitted on training data only. RF uses unscaled features.

The validation set is used for parameter selection, while the test set is reserved for final evaluation.

TABLE I
BEST CONFIGURATIONS ON CLEAN VALIDATION LABELS

Model	Selected Setting	Validation F1
LR	$k = 16,\ C = 10$	0.9312
RF	$k = 16$, depth 12, leaf 1	0.9317
MLP	$k = 16,\ \alpha = 0.01$	0.9393

III. Related Work

Zhang et al. [1] motivate the randomized-label experiment and demonstrate that training accuracy alone does not guarantee generalization.

Frenay and Verleysen [4] review the effects of mislabeled training examples on classification.

Koklu and Ozkan [3] describe the image-derived Dry Bean measurements used in our experiments.

Our project extends these ideas to a controlled comparison of linear, tree-based, and neural approaches on tabular data.

IV. Implemented Models and Experimental Design

A. Clean-Label Tuning and Feature Selection

We compare three classification models:

- **Logistic Regression (LR):** a regularized linear baseline.
- **Random Forest (RF):** a nonlinear ensemble of decision trees.
- **MLP:** a neural network with hidden layers $(128, 64)$.

We evaluated 18 parameter combinations using clean training labels and validation macro-F1.

For LR, we tested $k \in \{10, 16\}$ and $C \in \{0.1, 1, 10\}$.

For RF, we tested $k \in \{10, 16\}$, maximum depth in $\{12, \text{None}\}$, and minimum leaf size in $\{1, 3\}$.

For MLP, we tested $k \in \{10, 16\}$, $\alpha \in \{0.0001, 0.01\}$, and a learning rate of 0.001.

Hyperparameters were selected using validation performance only, without using test scores.

The final LR uses a maximum of 800 iterations, RF uses 200 trees, and MLP uses a maximum of 180 iterations without early stopping.

These settings are consistent with our current notebook.

B. Controlled Training-Label Randomization

For each training observation, its label is independently selected for replacement with probability p.

The replacement label is sampled uniformly from the seven bean classes, including the original class.

Therefore, the expected proportion of labels that actually change is

$$E[\text{changed fraction}] = p\left(1 - \frac{1}{7}\right) = \frac{6p}{7}. \tag{1}$$

We use five randomization probabilities:

$$p \in \{0, 0.25, 0.50, 0.75, 1.00\}.$$

The experiment is repeated with three random seeds: 17, 29, and 43.

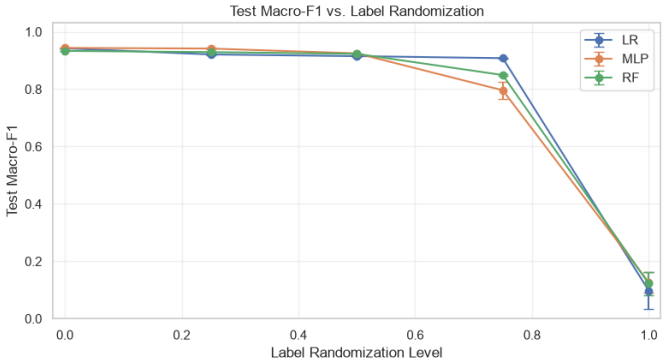


Fig. 2. Clean-test macro-F1 (mean ± SD) across three random seeds.

TABLE II
CLEAN-TEST MACRO-F1: MEAN ± SD

p	LR	RF	MLP
0.00	0.9423 ± 0.0000	0.9331 ± 0.0010	0.9436 ± 0.0011
0.25	0.9208 ± 0.0025	0.9289 ± 0.0022	0.9413 ± 0.0026
0.50	0.9149 ± 0.0026	0.9231 ± 0.0019	0.9246 ± 0.0007
0.75	0.9079 ± 0.0018	0.8490 ± 0.0035	0.7960 ± 0.0291
1.00	0.0976 ± 0.0637	0.1222 ± 0.0410	0.1264 ± 0.0350

For each seed and noise level, all three models use the same randomized training-label vector. Validation and test labels remain unchanged.

The dataset split is held fixed throughout the study.

This produces

$$3 \text{ models} \times 5 \text{ levels} \times 3 \text{ seeds} = 45 \text{ fits}.$$

Each fit records training accuracy on noisy and original labels, validation macro-F1, clean-test macro-F1, test accuracy, balanced accuracy, and runtime.

Means and standard deviations are reported across the three random seeds.

V. Preliminary Experimental Results

Our main comparison is shown in Fig. 2.

At $p = 0$, all three models achieve clean-test macro-F1 above 0.93.

Performance remains relatively stable through $p = 0.50$, but the models respond differently at higher randomization levels.

At $p = 0.75$, LR retains a macro-F1 of 0.9079, compared with 0.8490 for RF and 0.7960 for MLP.

At $p = 1.00$, performance decreases substantially for all three models.

A. Memorization and Generalization

RF maintains a relatively high noisy-label training accuracy at full randomization.

At $p = 1.00$, its average training accuracy on randomized labels is approximately 0.8461, while its clean-test macro-F1 is only 0.1222.

By comparison, LR and MLP achieve noisy-label training accuracies of 0.1597 and 0.2750.

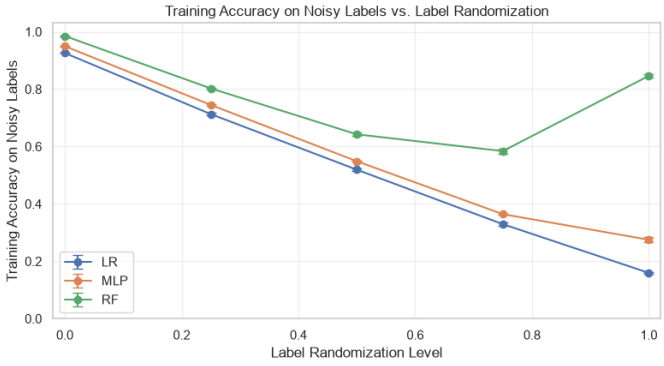


Fig. 3. Mean training accuracy measured against randomized training labels.

This suggests that RF is more capable of fitting randomized training labels under our settings, even though its generalization performance becomes poor.

The result supports the idea that fitting training labels does not necessarily mean learning useful patterns.

B. Initial Model Selection

For clean-label inference, MLP achieves the highest validation macro-F1 of 0.9393.

On the clean test set, the selected MLP achieves approximately 93% accuracy and 0.94 macro-F1.

Class-level evaluation shows that all 78 BOMBAY test observations are classified correctly.

SIRA is more difficult, with an F1-score of approximately 0.88.

The confusion matrix shows that 50 SIRA observations are predicted as DERMASON.

The selected model pipeline was saved using `joblib` and successfully predicted an example SEKER observation.

This confirms offline inference, while the web demonstration still requires final testing.

VI. Limitations, Roadmap, and Contributions

Limitations. Our experiment uses one dataset, one fixed data split, and three random seeds. Synthetic randomization may not represent real-world labeling errors. The MLP also has a limited training budget of 180 iterations. Future work could include additional datasets, longer training runs, and different label-noise mechanisms.

Team Contributions. Our responsibilities follow the project proposal. **Tianying Chen** focuses on coordination, data cleaning, EDA, and feature analysis. **Linjie Yang** focuses on model implementation, hyperparameter tuning, GitHub, and the web demo. **Elsey Chen** focuses on evaluation metrics, visualizations, model comparison, and interpretation. All three members will contribute to the final report and presentation.

Current Progress and Roadmap. We have completed the main machine learning code, including data preprocessing, three classification models, hyperparameter tuning, and 45 experimental fits. Preliminary analysis, result tables, and visualizations are also complete. Our next steps are to organize the GitHub repository, finish testing the web demonstration, further interpret the results, and prepare the final research paper and presentation.

References

[1] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, "Understanding deep learning requires rethinking generalization," in *International Conference on Learning Representations*, 2017, arXiv:1611.03530.

[2] UCI Machine Learning Repository, "Dry bean dataset," 2020. [Online]. Available: https://archive.ics.uci.edu/dataset/602/dry+bean+dataset

[3] M. Koklu and I. A. Ozkan, "Multiclass classification of dry beans using computer vision and machine learning techniques," *Computers and Electronics in Agriculture*, vol. 174, p. 105507, 2020.

[4] B. Frénay and M. Verleysen, "Classification in the presence of label noise: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 25, no. 5, pp. 845–869, 2014.